

Probabilistic Machine Learning — Project Presentation

Conditional DDPM with Classifier-Free Guidance

Controllable cartoon-face generation on Google's *Cartoon Set*

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From unconditional to controllable generation



Course baseline: unconditional DDPM

The model learns $p(x_0)$ and samples faces — but we cannot ask for a specific one.



Step 1 — attribute conditioning

The denoiser sees the labels: eye_color (5), hair_color (10), face_color (11).



Step 2 — classifier-free guidance (Ho & Salimans, 2022)

One network learns both conditional and unconditional models; a weight w amplifies the condition at sampling time.

32×32

resolution, RGB

10k → 100k

training images

≈4.4 M

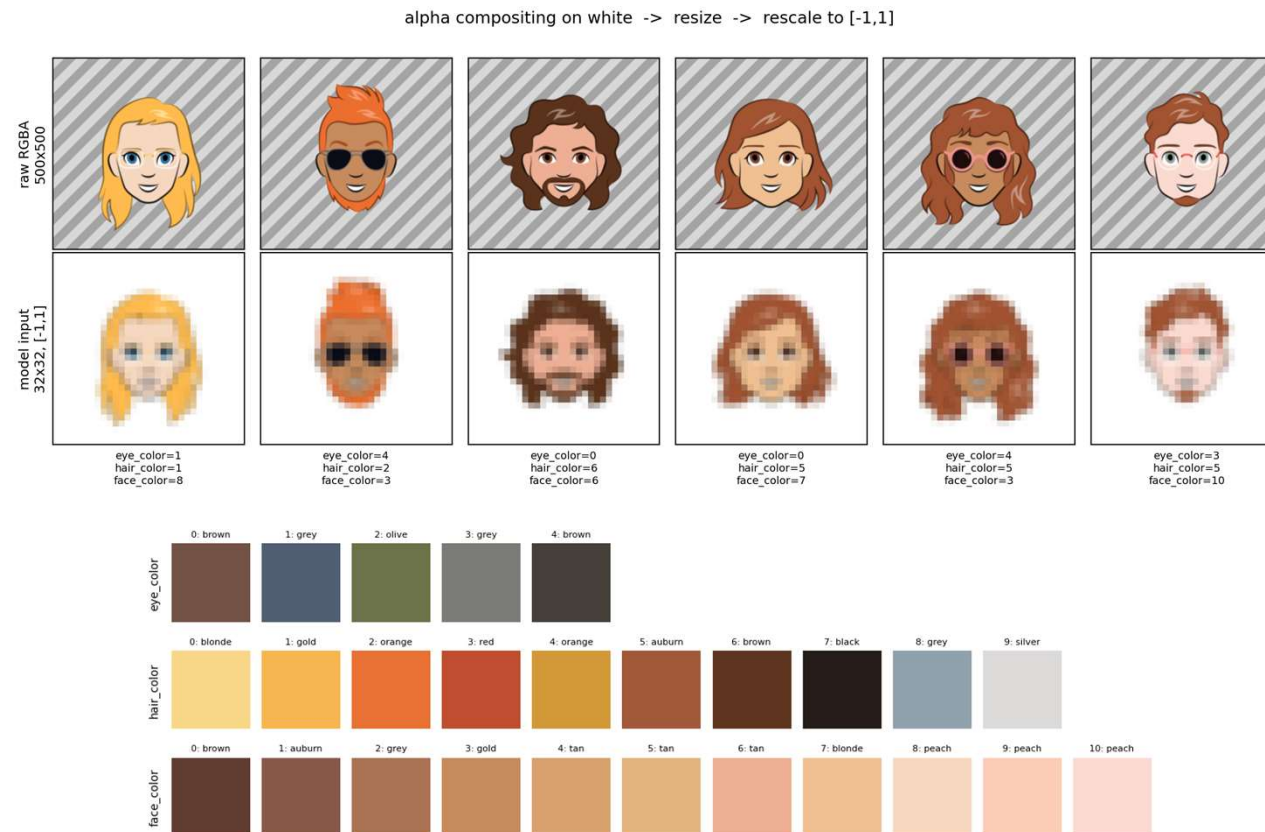
U-Net parameters

T = 1000

diffusion steps

The dataset

Visualization of the dataset we used: Google's Cartoon set



Conditional and Classifier-free guidance

What is guidance?

Noise prediction is score estimation — $\varepsilon_\theta(\mathbf{x}_t, t, c) \propto -\nabla \log q(\mathbf{x}_t | c)$ — so Bayes' rule splits the conditional score:

$$\nabla \log p(\mathbf{x}_t | c) = \nabla \log p(\mathbf{x}_t) + \nabla \log p(c | \mathbf{x}_t)$$

Conditioning alone

c enters the U-Net as one extra input: $\varepsilon_\theta(\mathbf{x}_t, t, c)$. The chain is untouched — but nothing controls how strongly samples express c .

Classifier guidance

Dhariwal & Nichol (2021): push samples along $w \cdot \nabla \log p(c | \mathbf{x}_t)$ from an external classifier — trained separately, on noisy images, at every t .

Classifier-free guidance

Ho & Salimans (2022): rewrite that term as:

$$\nabla \log p(c | \mathbf{x}_t) = \nabla \log p(\mathbf{x}_t | c) - \nabla \log p(\mathbf{x}_t)$$

two calls to one network, no classifier needed.

$$\tilde{\nabla} = \nabla \log p(\mathbf{x}_t | c) + w (\nabla \log p(\mathbf{x}_t | c) - \nabla \log p(\mathbf{x}_t))$$

Classifier-free guidance in one equation

One network, two models, a single knob w

$$\hat{\mathbf{e}} = (1 + w) \cdot \boldsymbol{\varepsilon}_{\theta}(\mathbf{x}_t, \mathbf{t}, \mathbf{c}) - w \cdot \boldsymbol{\varepsilon}_{\theta}(\mathbf{x}_t, \mathbf{t}, \emptyset)$$

Null token $\emptyset$

Each attribute embedding table has one extra row: the “unconditional” token used to query the same network without labels.

Condition dropout

During training the label vector is replaced by $\emptyset$ with probability $p = 0.1$, so a single network learns both models jointly.

Reading w

$w = 0$ is the plain conditional model. Larger w pushes samples where the attribute is more strongly expressed — fidelity up, until it isn't.

Pipeline and implementation



dataset.py

Alpha compositing on white, resize 32×32, $[-1,1]$ scaling; attribute cardinalities parsed from data, never hard-coded.



diffusion.py

Linear β schedule, q_{sample} , training loss with condition dropout, guided DDPM & DDIM samplers.



sample.py

Grids: attribute values × guidance weights, same starting noise per row — isolates the effect of w .



model.py

Conditional U-Net



train.py

Adam ($\text{lr } 2 \cdot 10^{-4}$), EMA 0.999, collision-proof checkpoints with .bak rotation.



evaluate.py

Independent multi-head CNN attribute classifier trained on real images → fidelity and diversity vs w .

Measuring what guidance does

A generator cannot grade its own homework



Attribute classifier

Small multi-head CNN trained on real images only — an independent measuring instrument, cached to disk.



Attribute fidelity vs w

Generate from random attribute vectors, classify, report the fraction matching the request — per attribute and mean.



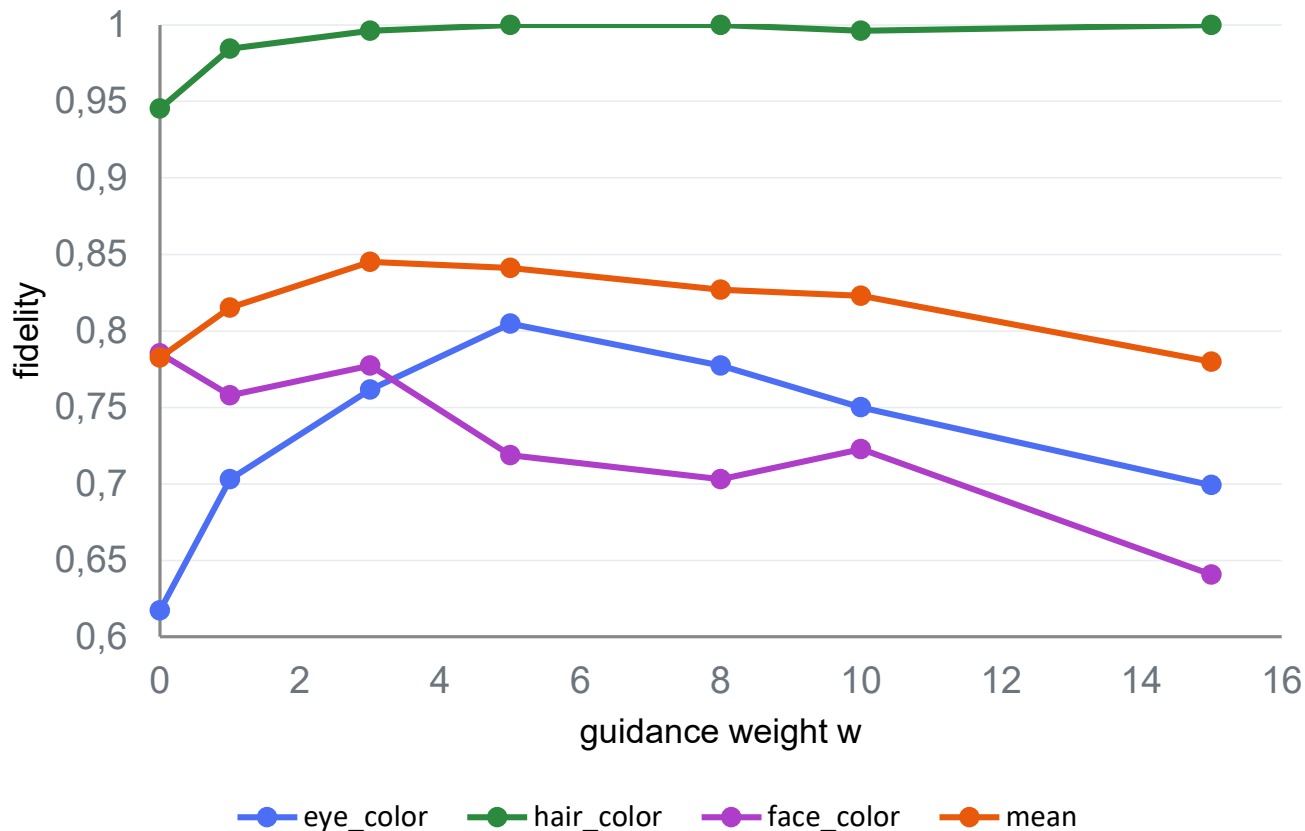
Diversity check

Mean pairwise distance between same-condition samples: does high w trade variety for fidelity (mode collapse)?

Sweep: $w \in \{0, 1, 3, 5, 8, 10, 15\}$ · full ancestral DDPM sampling (1000 steps)

Attribute fidelity vs guidance weight

10k images · 40k steps · DDPM sampler · $w \in \{0, \dots, 15\}$



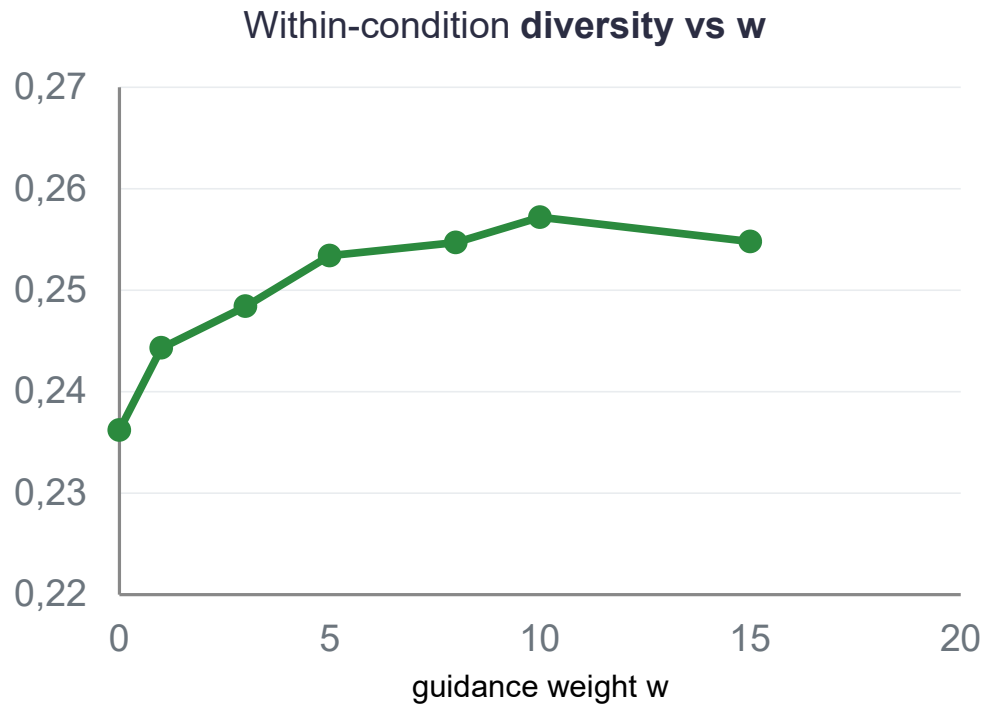
0.845

peak mean fidelity at $w = 3$
(vs 0.783 at $w = 0$)

- hair_color saturates at 1.0 for $w \geq 3$ (ceiling effect)
- face_color degrades first — oversaturation pushes global colour out of distribution
- gentle decline by $w = 15$: a genuine CFG ceiling, not a sampler artifact

No fidelity–diversity trade-off in this range

Same sweep — within-condition diversity and run-to-run variability



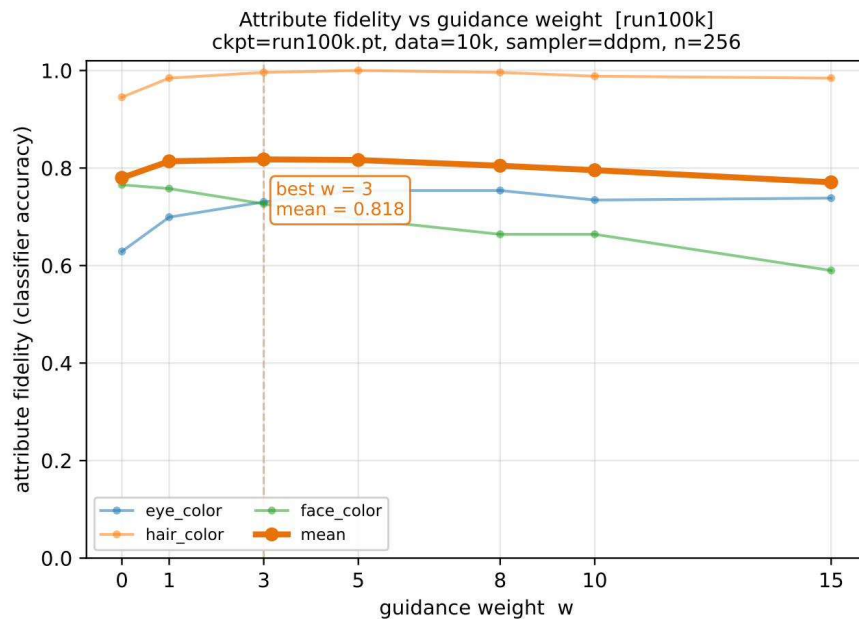
Diversity does not collapse — it rises mildly ($0.236 \rightarrow 0.257$): guidance is not buying fidelity by repeating the same image. **Run-to-run $\sigma \leq 0.017$: the fidelity peak at $w \approx 3$ is well outside noise.**

Scaling the data: the 100k run

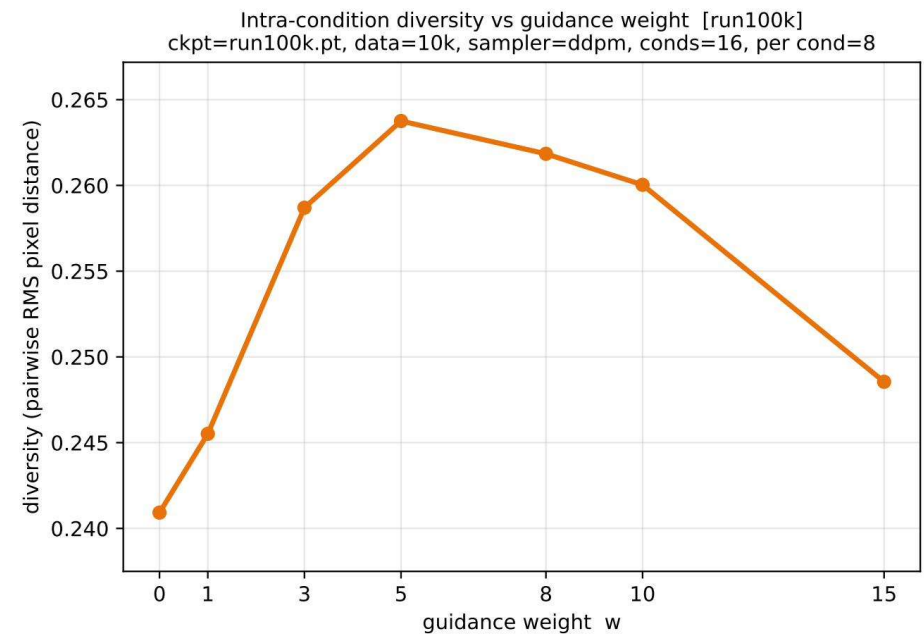
40k steps on 10k images $\approx$ 512 epochs — more data beats more steps

dataset100k.py — recursive scan of nested folders, one-time atomic disk cache ($\approx$ 307 MB at 32px), corruption-safe.

train100k.py — resumable (--resume), step/s + ETA logging. Same 40k steps $\rightarrow$ $\approx$ 51 epochs: each image seen 10 $\times$ less often.



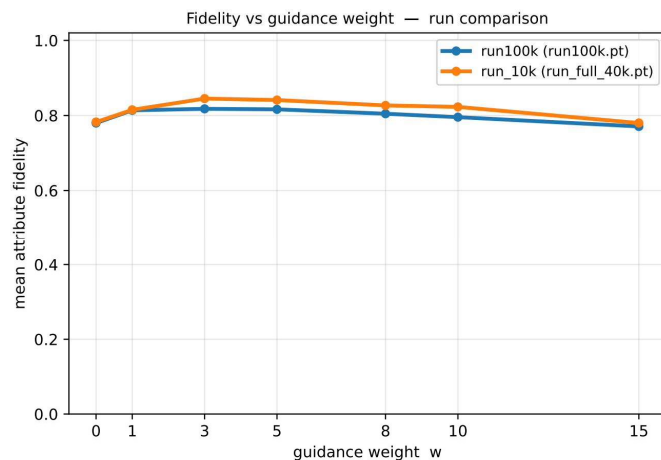
run100k.pt · same sweep as the 10k checkpoint



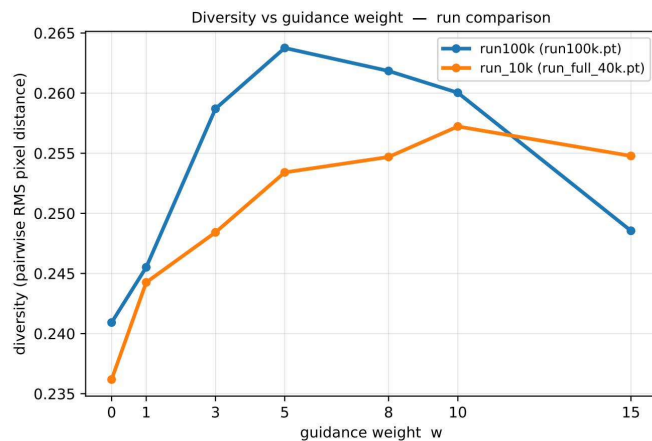
Diversity peaks near w = 5 then eases — no collapse at high w

10k vs 100k: does more data help?

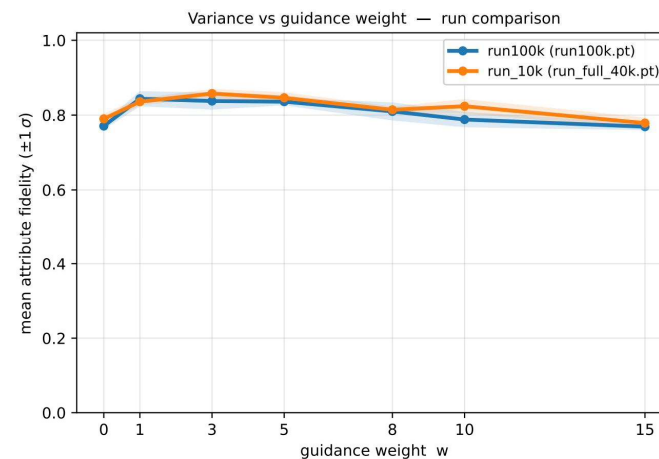
Same architecture, same 40k steps, same sweep — only the dataset changes



Fidelity: 10k edges out 100k at every w



Diversity: 100k is higher — more data, more variety

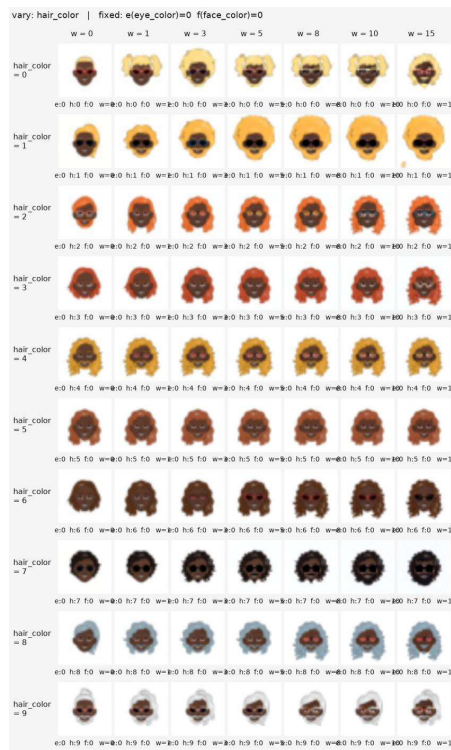


Both peak at $w \approx 1-3$, within each other's noise band

More data raises diversity, not fidelity, at this scale. At 40k steps the 100k run has seen each image 10× less often (≈ 51 vs ≈ 512 epochs), so its mean fidelity (0.818 peak) trails the 10k run (0.845 peak) by a small, consistent margin at every w. The optimal guidance weight itself ($w \approx 3$) does not shift with scale — it is a property of the CFG mechanism, not of dataset size.

Seeing the guidance effect

sample.py — same starting noise per row, only w changes across columns



Grid: $hair_color \times w \cdot w \in \{0, 1, 3, 5, 8, 10, 15\}$



Grid: $face_color \times w \cdot$ oversaturation at high w

Because the noise is fixed per row, every visible change is attributable to w alone.

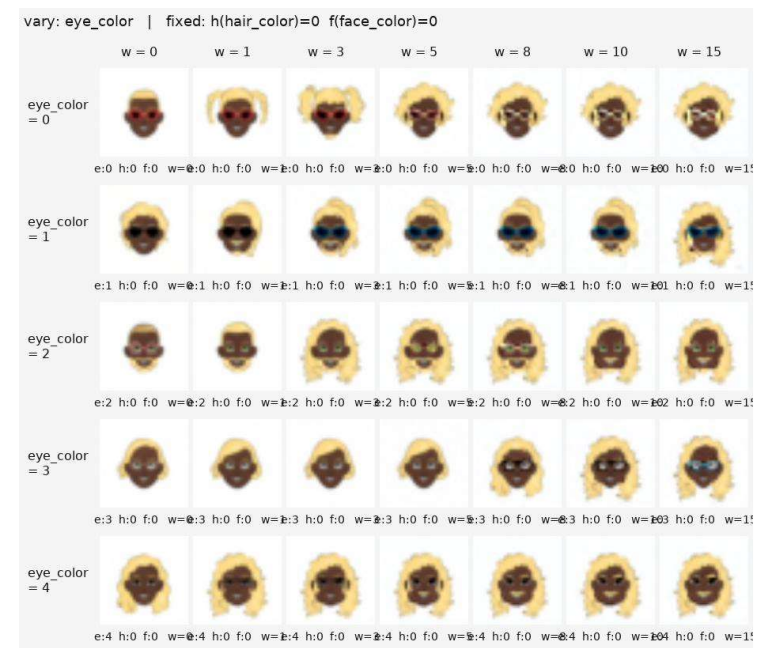
eye_color across scale: 10k vs 100k

Same fixed noise per cell, same sweep — only the training set differs

10k images — ≈512 epochs



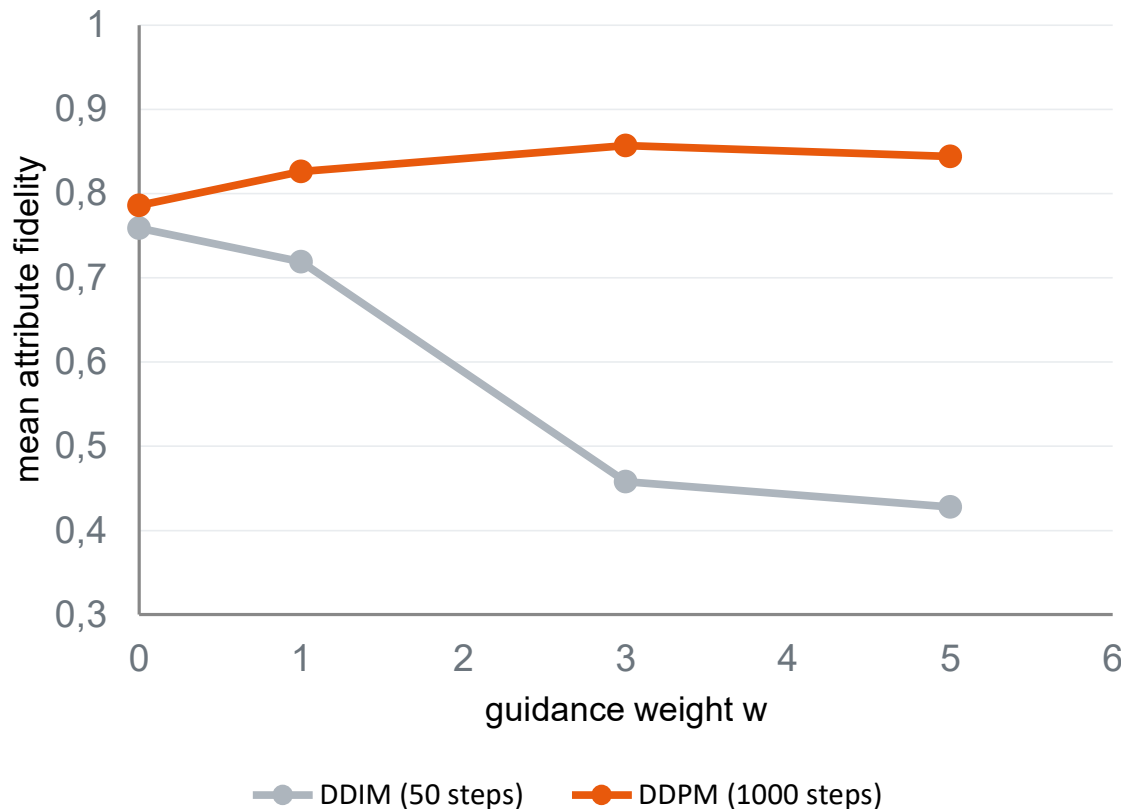
100k images — ≈51 epochs



Qualitatively the two checkpoints are hard to tell apart — both track eye_color cleanly and sharpen with w. That mirrors the numbers: eye_color fidelity stays close at every w (e.g. 0.80 vs 0.75 at w = 5), consistent with the 100k run's small, uniform fidelity deficit rather than a qualitatively different failure mode.

A sampler artifact that flips the conclusion

Same checkpoint, same guided noise — opposite result



Under DDIM, fidelity collapses for $w > 1$

0.76 $\rightarrow$ 0.43 — the opposite of what CFG should do, and inconsistent with the clean qualitative grids. 250 steps did not fix it.



Diagnosis

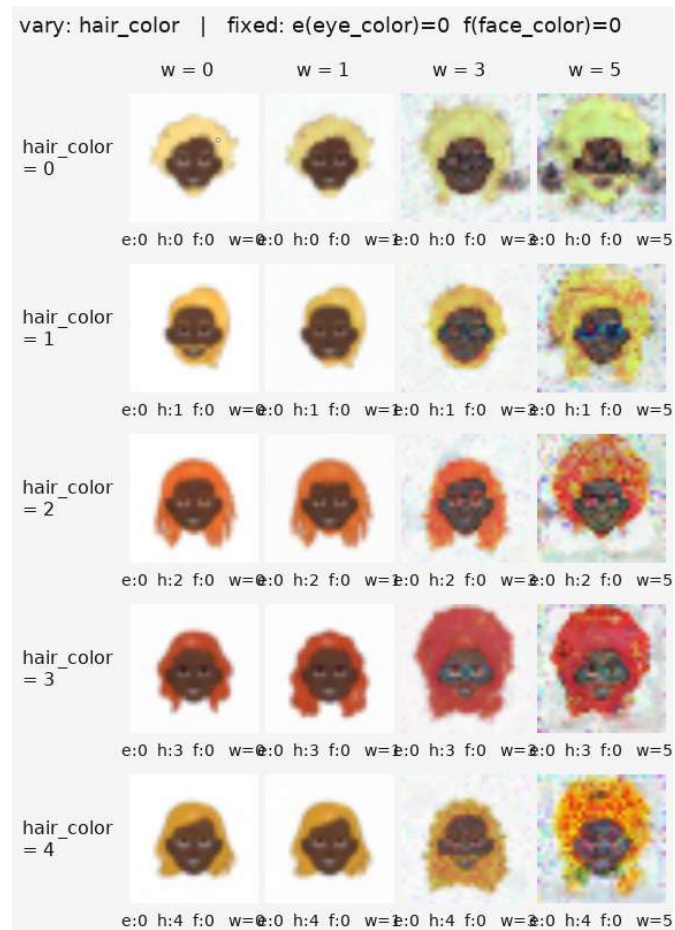
DDIM reconstructs $\hat{x}_0$ explicitly each step; at high t the coefficient is huge (≈ 157 at $t = 999$), amplifying the off-manifold guided noise. DDPM never forms $\hat{x}_0$ and takes 1000 small steps.



Decision

All reported results use full ancestral DDPM. Both evaluators expose `--sampler {ddim, ddpm}` for reproducibility.

What happens with DDIM?



Takeaways

1

One network, two models

The null token + 10% condition dropout make CFG almost free at training time — the cost is one extra forward pass per sampling step.

2

Guidance works — in moderation

Mean fidelity 0.783 $\rightarrow$ 0.845 at $w = 3$, with no diversity collapse; colour attributes saturate and degrade gently past $w \approx 5$.

3

Guidance helps — with some drawbacks

DDIM under guidance produced a spurious collapse that would have inverted the conclusion. Evaluation choices deserve the same scrutiny as the model.

Thank you — questions welcome